

AI Learning Hub – Project Proposal

AI-Powered Personalized Learning Platform

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1. Executive Summary

Ai Learning Hub is an AI-powered personalized learning platform designed to improve how learners discover and progress through online courses. It uses a hybrid recommendation system combining content-based filtering, collaborative filtering, and re-ranking techniques powered by open-source HuggingFace models.

Unlike traditional e-learning systems that provide static recommendations, this platform adapts dynamically to user interests, behavior, and career goals. The system is built using a modern full-stack architecture (FastAPI, React, SQLite) and runs entirely on free, open-source technologies, making it suitable for both academic and practical use.

2. Problem Statement

With the rapid growth of online learning platforms, learners face several challenges:

Information Overload: Too many courses make it difficult to choose relevant content.

Lack of Personalization: Recommendations are often generic and not tailored to individual needs.

No Learning Path Guidance: Platforms fail to provide structured progression toward career goals.

These issues lead to reduced engagement and inefficient learning.

3. Objectives

Primary Objectives

Develop a complete AI-based personalized recommendation system.

Implement content-based, collaborative, and hybrid recommendation models.

Build a full-stack application integrating ML models with a user interface.

Secondary Objectives

Use only open-source HuggingFace models for all AI functionalities.

Provide an AI chatbot for course guidance and learning assistance.

Generate personalized learning paths based on user goals and progress.

4. Project Scope

In Scope

Full-stack web application

Recommendation system (3 models)

AI chatbot

Progress tracking and learning path generation

Local ML experiment tracking

Payment systems

Out of Scope

Cloud deployment

Mobile applications

CI/CD pipelines

5. System Architecture

The system follows a layered architecture:

Frontend: React-based UI for user interaction

Backend: FastAPI handling APIs and business logic

Database: SQLite for storing users, courses, and interactions

ML Layer: Recommendation models and embeddings

MLOps: MLflow for tracking model performance

Flow:

User → Frontend → Backend → ML Models → Recommendations → UI

6. Methodology

Data Preparation

Synthetic dataset with users, courses, and interactions

Recommendation System

Content-Based Filtering: Matches user interests with course descriptions

Collaborative Filtering: Suggests courses based on similar users

Hybrid Model: Combines both approaches with re-ranking

AI Features

Search: Semantic search using embeddings

Chatbot: Conversational assistant for course guidance

Auto Tagging: Zero-shot classification for course categories

7. Technology Stack

Backend: Python, FastAPI

Frontend: React, Tailwind CSS

Database: SQLite

Machine Learning: HuggingFace Transformers, Sentence Transformers

Libraries: NumPy, scikit-learn, implicit

MLOps: MLflow

8. Deliverables

Fully functional web application

Recommendation system implementation

AI chatbot integration

Synthetic dataset generator

Documentation and source code

9. Expected Outcomes

Technical

Working AI-powered recommendation system

Efficient and scalable architecture

Measurable improvement using evaluation metrics

User Experience

Personalized course recommendations

Structured learning paths

Interactive AI assistance

10. Conclusion

Ai learning Hub provides an effective solution to modern e-learning challenges by integrating AI-driven personalization with a scalable architecture. The use of open-source models ensures accessibility, while the hybrid recommendation approach improves learning efficiency and user engagement.

This project demonstrates the practical application of AI in education and serves as a strong foundation for future research and development